

Mathematics II

BCA — Semester II — 2023 — Answer Sheet
Subject Code: CACS154

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Evaluate the limit $\lim_{x \rightarrow \theta} \frac{x \cos \theta - \theta \cos x}{x - \theta}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Find the derivative of $y = e^{3x - 1}$ by the definition method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Verify Rolle's theorem for the function $f(x) = \sin x, x \in [0, \pi]$ and find the point on the curve where the tangent is parallel to the x-axis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find maximum and minimum values of the function $f(x) = x^3 + 6x^2 + 9x - 2$. Also find the point of inflection, if any.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Evaluate the integrals. a. $\int \frac{2x + 5}{\sqrt{x^2 + 5x}} dx$ b. $\int e^{2x} \log x dx$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Evaluate $\int_0^2 \sqrt{1 + x^3} dx$ by using Simpson's $\frac{1}{3}$ rule, taking $(n = 4)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the differential equation: $2 \frac{dy}{dx} = \frac{2y}{x} + \frac{y^2}{x^2}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions [2x10=20]

9. Define pivot element, pivot column, and consistency in a system of equations. Using the simplex method, maximize $F = 5x - 3y$ subject to $\begin{cases} 3x + 2y \leq 6, \\ -x + 3y \geq -4, \\ x \geq 0, y \geq 0. \end{cases}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. a. Verify Lagrange's mean value theorem for the function $f(x) = \sqrt{x - 1}$, $x \in [1, 3]$. b. Solve the differential equation $x \frac{dy}{dx} = x^2 + y^2$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Compute the approximate value of the integrate $\int \frac{1}{1 + x^2} dx$ by using the composite trapezoidal rule with three points, and compare the result with the actual value. Determine the error formula and numerically verify an upper bound on it.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.